

Q21. Application: Autonomous Vehicle Navigation System

Analysis of Autonomous Vehicle Navigation System Using Master Theorem

Problem Statement

An autonomous vehicle navigation system processes route data using the recurrence relation:

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

Apply the **Master Theorem** to determine the time complexity. Clearly identify the parameters, justify the selected case, analyze the effect of increasing map data on performance, and interpret the efficiency using asymptotic notation.

1. Problem Context

In an autonomous vehicle navigation system, the navigation algorithm divides the map into two equal parts and processes each part recursively. After processing the subproblems, additional computations such as obstacle detection, route optimization, and traffic analysis require n^2 time.

The recurrence relation representing the algorithm is:

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

The objective is to determine the algorithm's time complexity using the **Master Theorem**.

2. Application of Master Theorem

The standard form of the Master Theorem is:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Identify the Parameters

- $a = 2$ (Number of recursive calls)
- $b = 2$ (Input size divided by 2)
- $f(n) = n^2$

Now calculate:

$$\begin{aligned} n^{\log_b a} \\ &= n^{\log_2 2} \\ &= n^1 = n \end{aligned}$$

Thus,

$$n^{\log_b a} = n$$

Compare:

$$f(n) = n^2$$

Since

$$n^2 = \Omega(n^{1+\varepsilon})$$

where

$$\varepsilon = 1$$

the function $f(n)$ grows polynomially faster than $n^{\log_b a}$.

3. Selection of Master Theorem Case

The three cases of the Master Theorem are:

- **Case 1:** $f(n)$ is smaller
- **Case 2:** $f(n)$ is equal
- **Case 3:** $f(n)$ is larger

Since

$$f(n) = n^2$$

is asymptotically larger than

$$n^{\log_b a} = n,$$

Case 3 is applicable.

The regularity condition is verified:

$$af\left(\frac{n}{b}\right) = 2\left(\frac{n}{2}\right)^2 = 2 \cdot \frac{n^2}{4} = \frac{n^2}{2}$$

Since

$$\frac{n^2}{2} \leq cn^2$$

for some constant $c < 1$ (for example, $c = \frac{1}{2}$), the regularity condition is satisfied.

Therefore,

$$\boxed{T(n) = \Theta(n^2)}$$

4. Time Complexity

Using **Master Theorem Case 3**,

$$T(n) = \Theta(n^2)$$

Hence, the time complexity of the autonomous navigation algorithm is **Quadratic Time**.

5. Performance Analysis

As the map data increases, the execution time grows quadratically.

For example:

Number of Map Nodes (n)	Approximate Operations (n ²)
100	10,000
500	250,000
1,000	1,000,000
2,000	4,000,000

This shows that when the number of map nodes doubles, the computation increases by approximately four times.

Therefore:

- Small maps are processed efficiently.
 - Large city-wide maps require significantly more computation.
 - Processing time increases rapidly as the amount of route data grows.
 - Additional optimization techniques or parallel processing may be required for real-time autonomous navigation.
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6. Interpretation Using Asymptotic Notation

The asymptotic complexity is

$$\boxed{\Theta(n^2)}$$

This means the algorithm has a quadratic growth rate.

- **Big-O:** $O(n^2)$ – Upper bound
- **Big-Omega:** $\Omega(n^2)$ – Lower bound
- **Big-Theta:** $\Theta(n^2)$ – Tight bound

Thus, the algorithm consistently requires quadratic time for large input sizes.

7. Final Conclusion

The autonomous vehicle navigation system follows the recurrence relation

$$T(n) = 2T(n/2) + n^2.$$

Using the Master Theorem:

- $a = 2$
- $b = 2$
- $f(n) = n^2$
- $n^{\log_b a} = n$

Since n^2 grows faster than n , **Master Theorem Case 3** applies. The regularity condition is satisfied, resulting in the final time complexity:

$$\boxed{T(n) = \Theta(n^2)}$$

This indicates that the algorithm's running time increases quadratically as the size of the map data grows. While suitable for small and medium-sized maps,

processing very large navigation datasets becomes computationally expensive, making optimization techniques important for real-time autonomous vehicle systems.